

Nuclear Fuel Performance

NE-533
Spring 2026

Last Time

- Developed equations for stresses in a thick-walled cylinder
- Thermal expansion causes strain, but doesn't necessarily cause stress
- The thermal gradient in the fuel and cladding lead to stresses
- Developed equations for stresses due to thermal expansion in a cladding tube, and then in the fuel
- The thermal stresses can induce radial cracks in the fuel due to high hoop stress
- Change in gap thickness due to thermal expansion

NUMERICAL THERMO-MECHANICS

Now we can solve the temperature and the displacement vector for the full thermomechanical problem

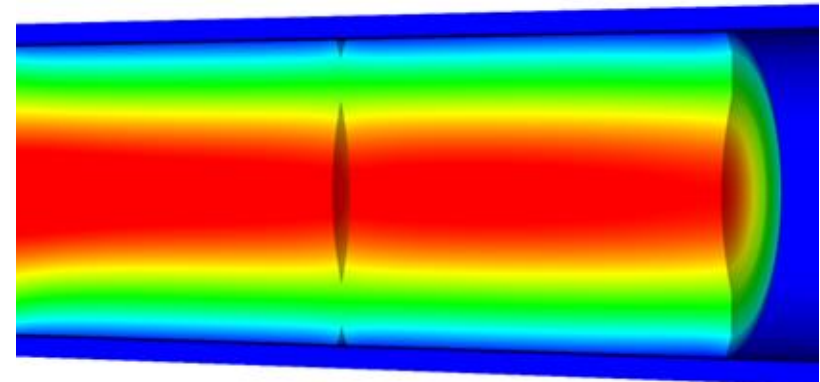
- T impacts the value of $\mathbf{u}$ through thermal expansion
- $\mathbf{u}$ impacts the value of T through changes in the thickness of the gap
- The value for T evolves with time
- The value for $\mathbf{u}$ also evolves with time, even though there is not time in its PDE

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q$$

$$\boldsymbol{\sigma} = \mathcal{C}(\boldsymbol{\epsilon} - \alpha(T - T_{fab})\mathbf{I})$$

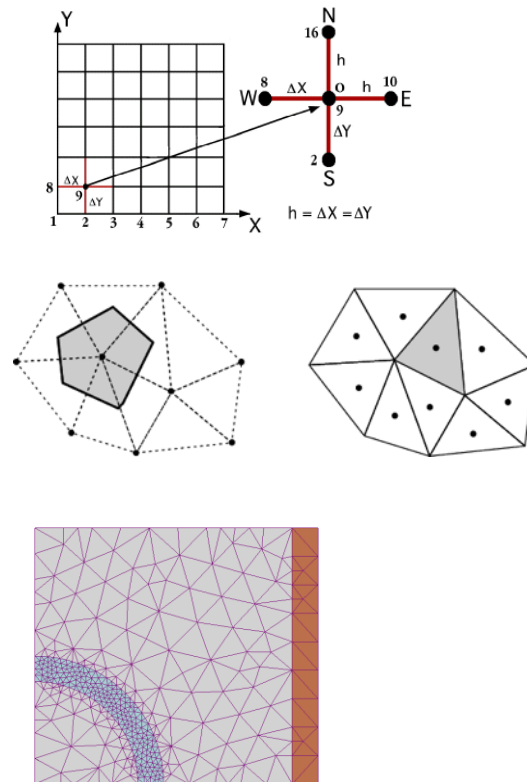
$$0 = \nabla \cdot \boldsymbol{\sigma}$$

$$\boldsymbol{\epsilon} = \frac{1}{2} (\nabla \mathbf{u} + \nabla \mathbf{u}^T)$$

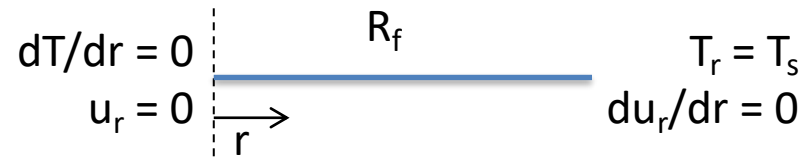


The primary tool for solving all thermomechanics problems is the finite element method

- **Finite difference**
 - Can solve the heat conduction equation
 - Can't easily solve the mechanics equations
- **Finite Volume**
 - Can solve the heat conduction equation
 - Can solve the mechanics equations
 - Eqns derived from fluid dynamics
- **Finite Element**
 - Can solve the heat conduction equation
 - Can solve the mechanics equations
 - Can handle any geometry
 - Can handle any boundary condition



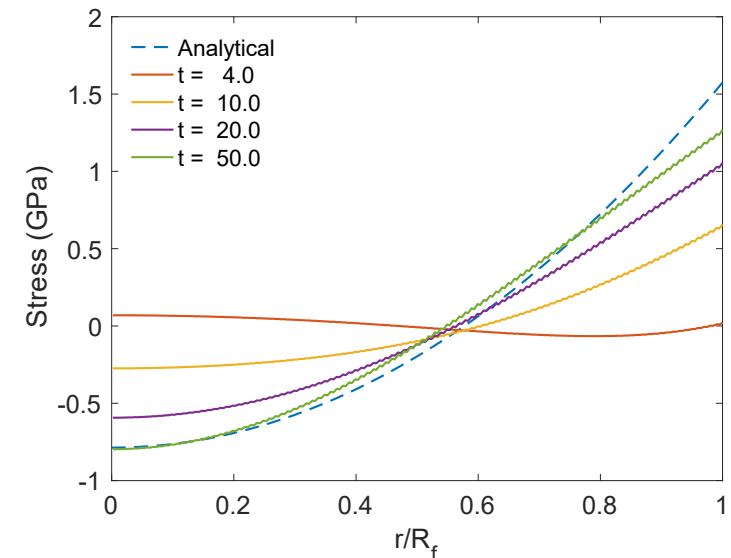
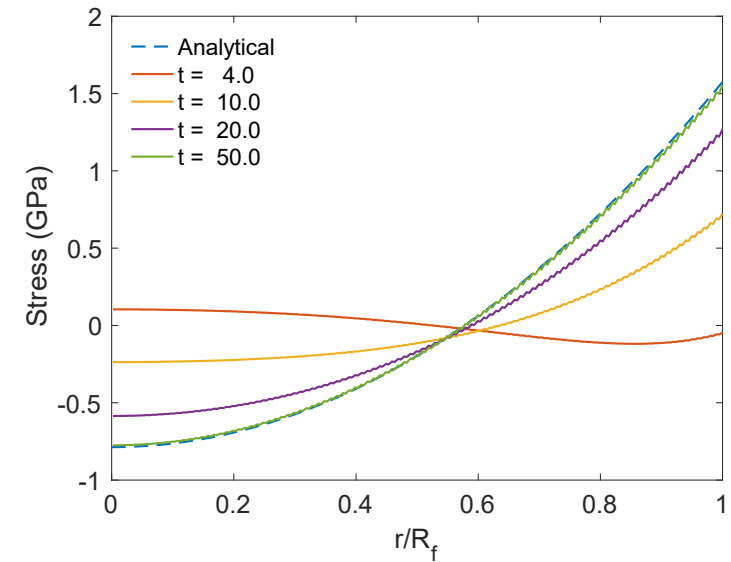
The 1D thermomechanics problem definition


$$\begin{array}{ccc} dT/dr = 0 & R_f & T_r = T_s \\ u_r = 0 & \xrightarrow{r} & du_r/dr = 0 \end{array}$$

- The initial temperature is set to 273 K
- We will take 50 time steps of 0.5 s
- The full power of $Q = 450$ begins at time $t = 0$.
- UO_2 material properties are used for both the thermal and mechanics equations

Comparison to analytical theory

- If we use a constant thermal conductivity, analytical 1D model matches very well
- When k is a function of temperature, there is a difference between the FEM and analytical stress



There are various available tools for solving coupled thermomechanical problems with FEM

- Commercial tools
 - ABAQUS
 - ANSYS
 - COMSOL
- Open source
 - MOOSE
- NRC-based
 - FRAPCON/FRAPTRAN

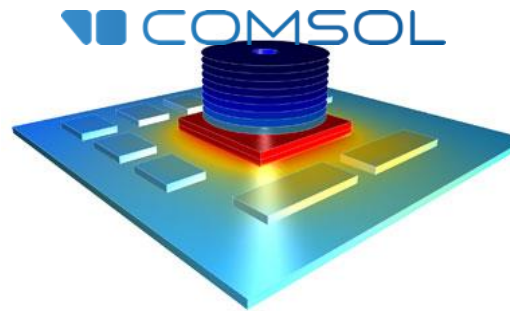
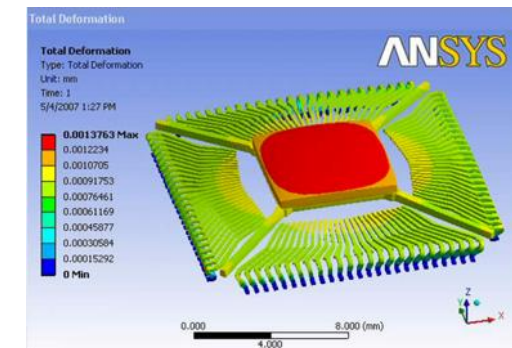
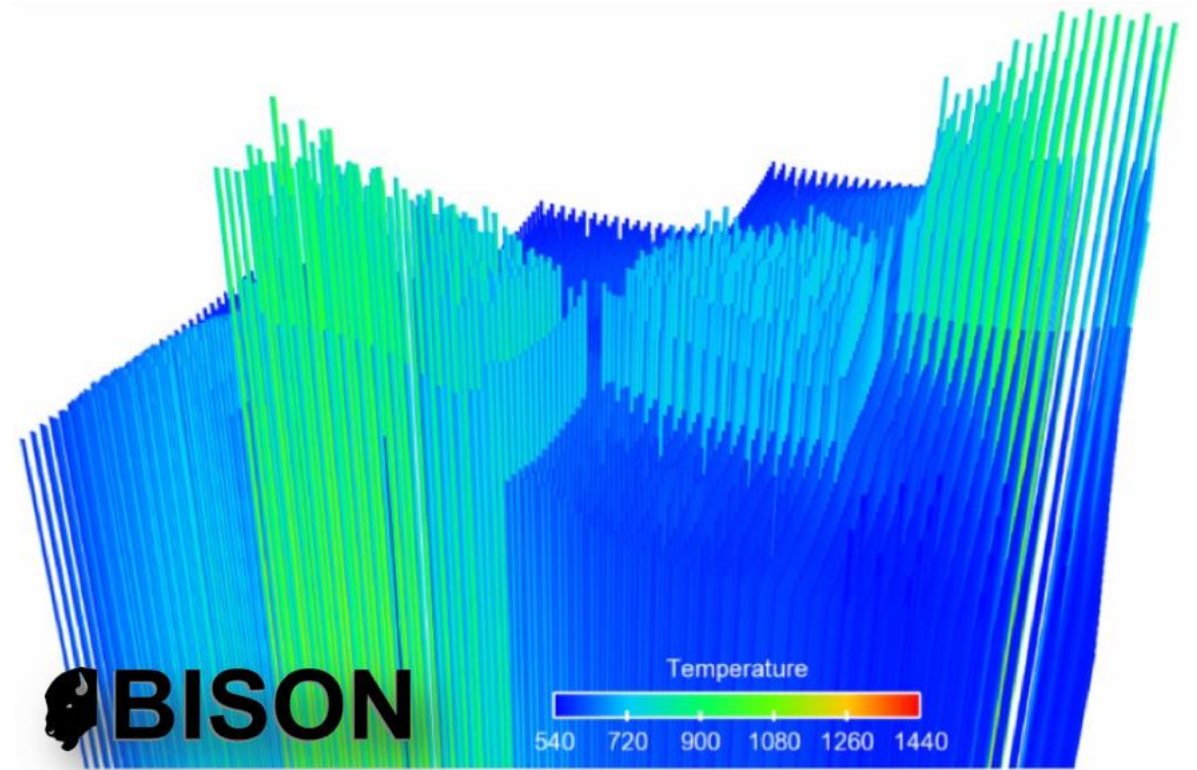


Figure 4. Temperature contour plot of exhaust manifold.



The purpose of a fuel performance code is to simulate and evaluate fuel rod behavior

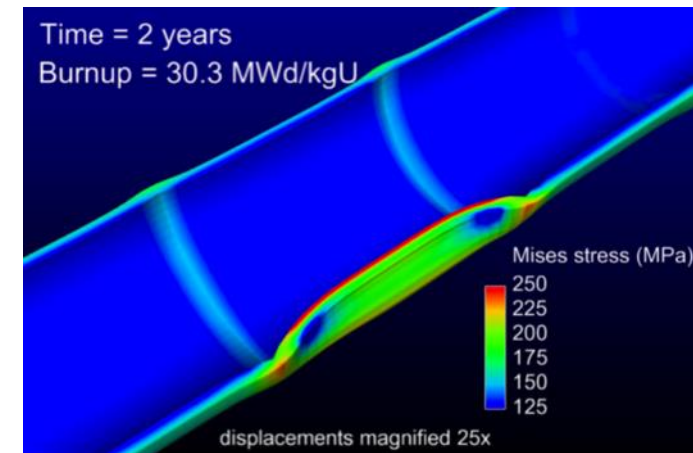
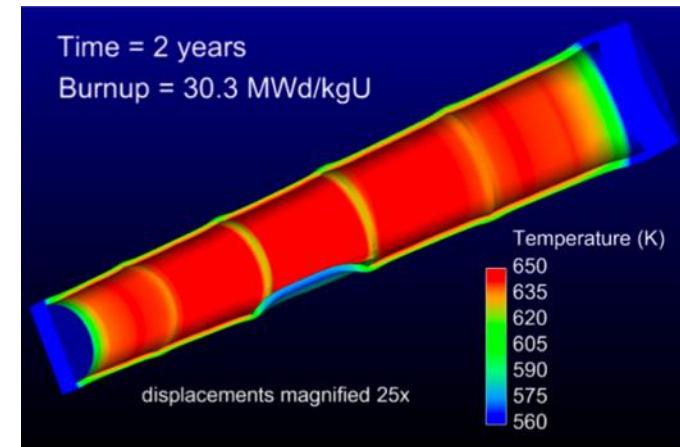
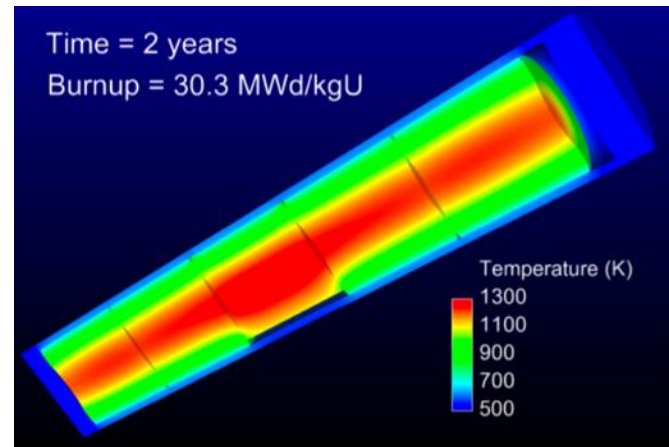
- The first fuel performance codes were developed in the mid seventies
- Advanced fuel performance codes are still under development today



KAIST-3A benchmark results, showing displacement of 3432 rods, from Gaston et al. 2014

How are fuel performance codes used?

- The primary goals are to predict the fuel centerline temperature and the stress in the cladding
- Fuel performance codes aren't focused on predicting power production, but rather to predict safety margins, provide coupling to other codes



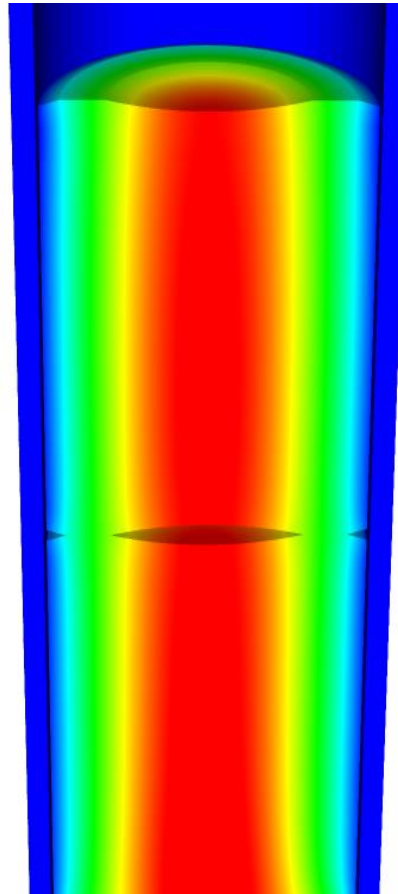
A fuel performance code must be able to predict:

Fuel

- Temperature profile
- Volumetric change

Cladding

- Temperature profile
- Stress



Gap

- Gap heat transport
- Mechanical interaction between fuel and cladding
 - Gap pressure

The primary focus is solving the thermomechanical problem

Fuel

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q$$

Solved Numerically

$$0 = \nabla \cdot \sigma$$

*Solved Numerically or
analytically*

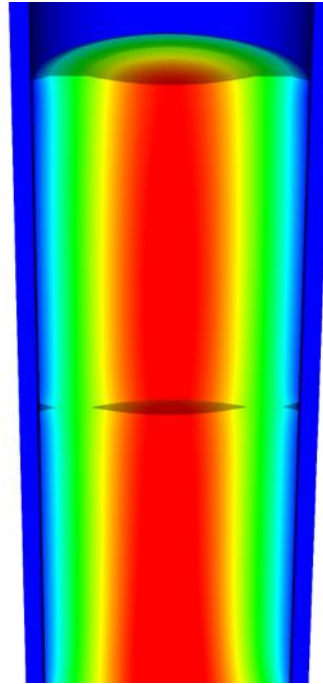
Cladding

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q$$

*Solved Numerically or
analytically*

$$0 = \nabla \cdot \sigma$$

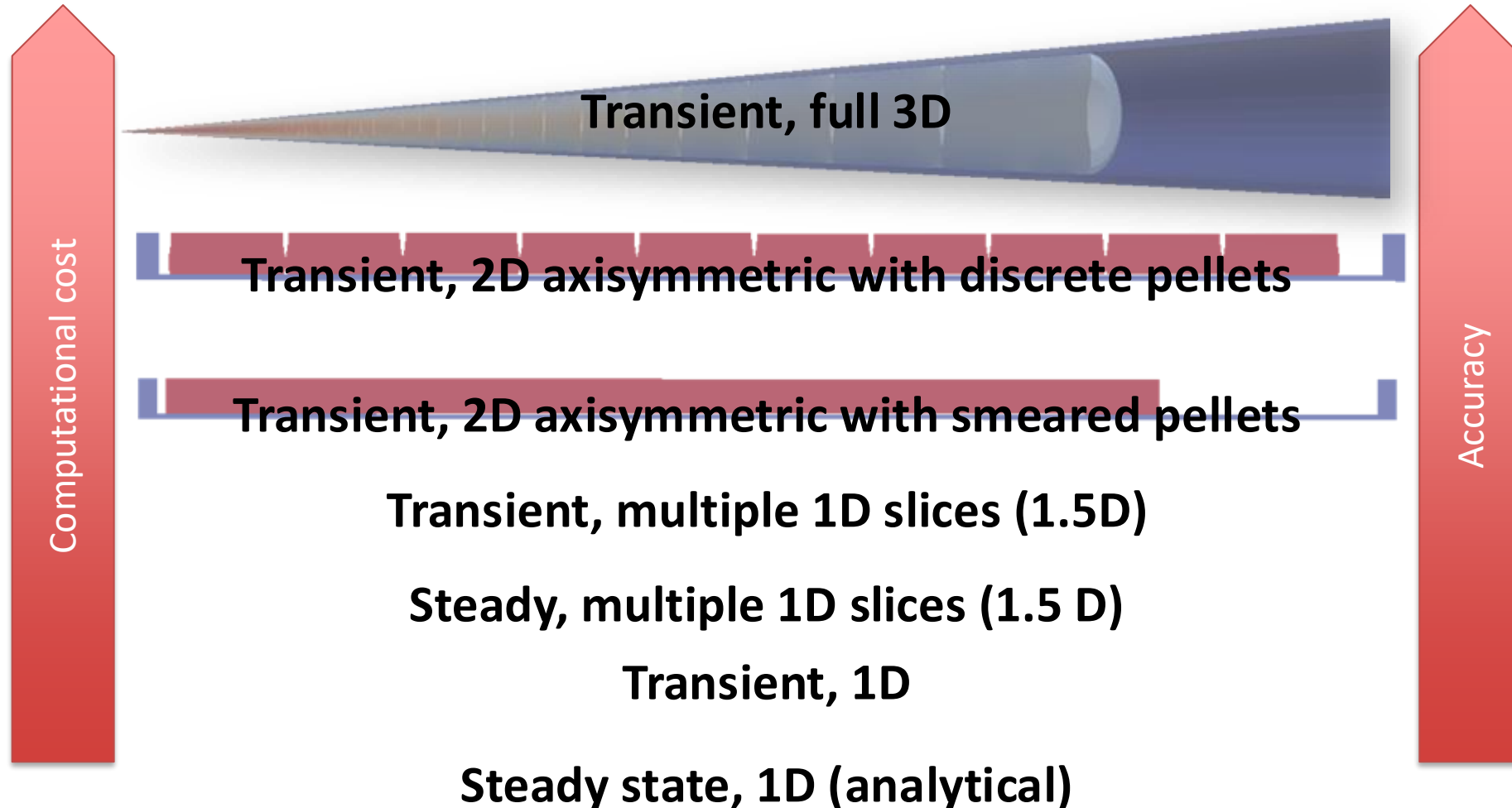
Solved Numerically



Gap

- The handling of the gap changes the most between different codes

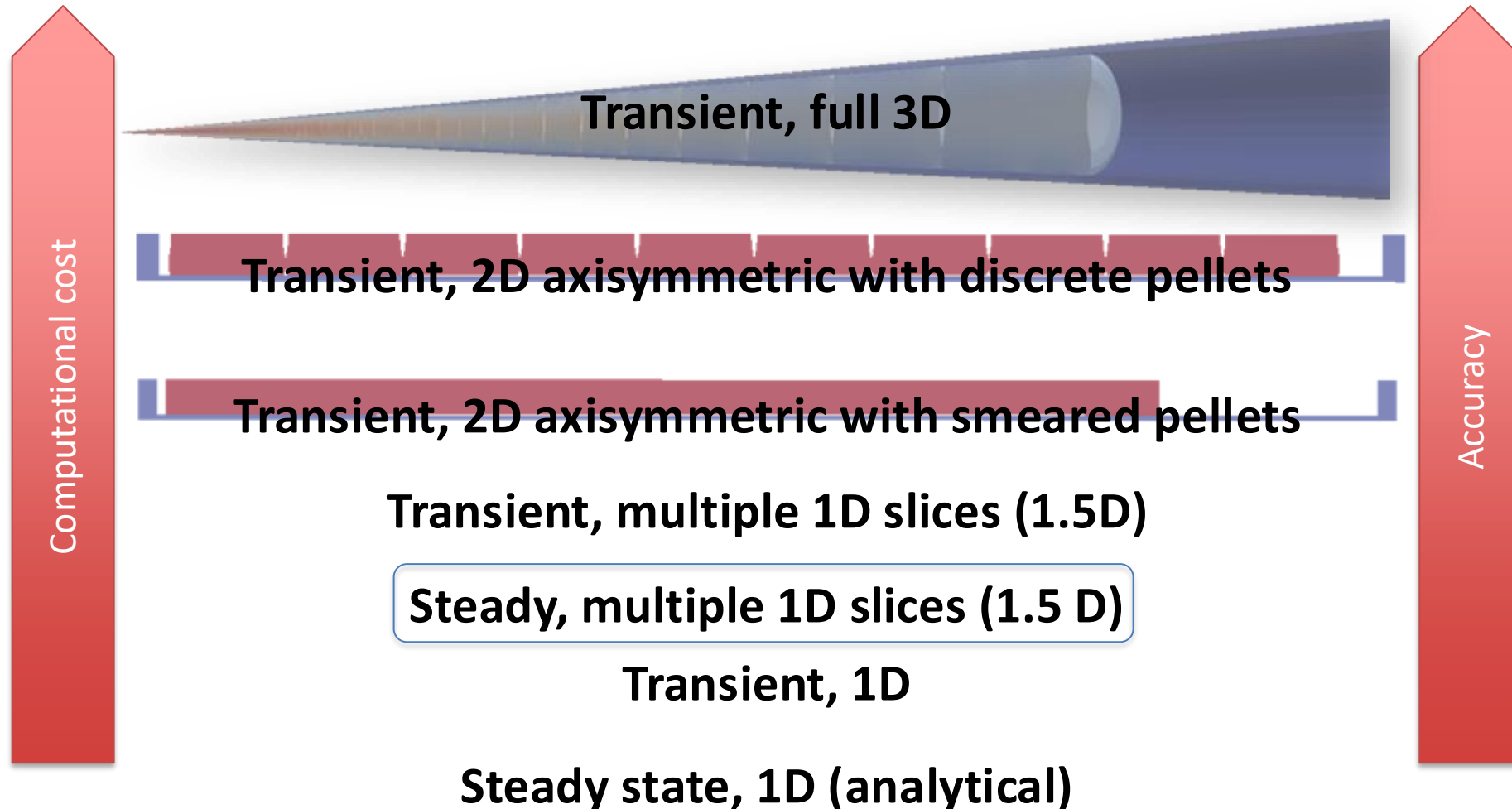
Various approaches to model the fuel rod



Early fuel performance codes were made for either steady state or transient operation

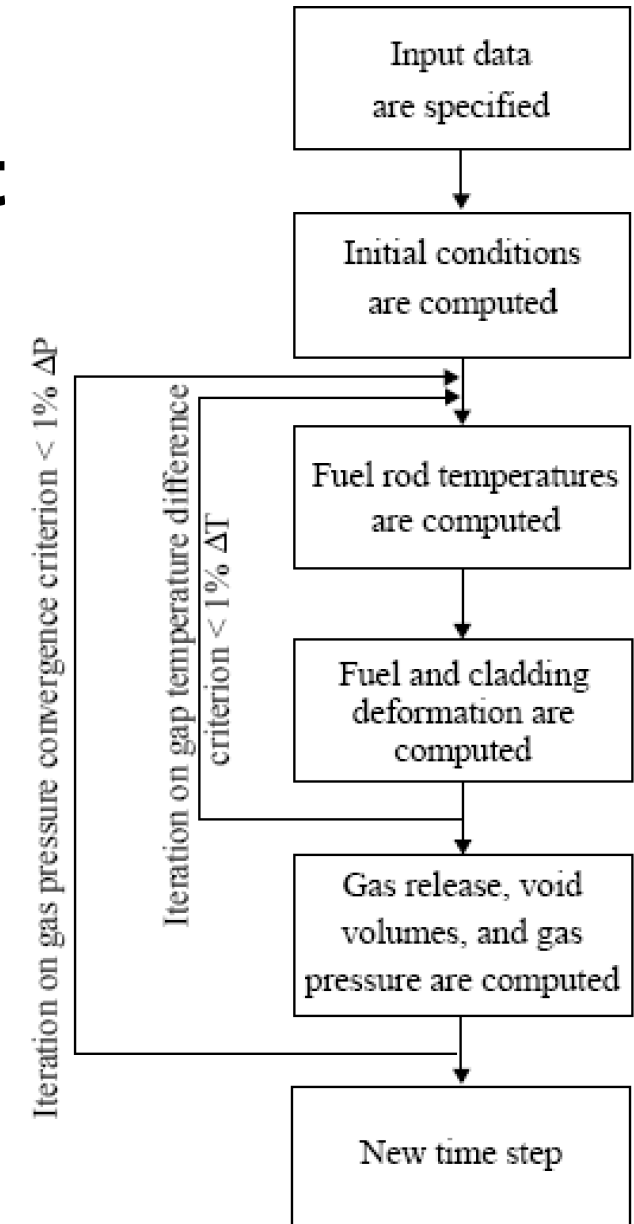
- Steady state codes
 - Leave off the time derivative part of the heat equation $\nabla \cdot (k \nabla T) + Q = 0$
 - The material properties still evolve with time as a function of burnup
 - The volumetric changes in the fuel are also a function of burnup
 - Creep of fuel and cladding change with time
- Transient codes
 - Include the time derivative $\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q$
 - Have similar burnup dependent models like steady state codes, but don't include creep
 - Have additional models for rapid transients

Start with FRAPCON

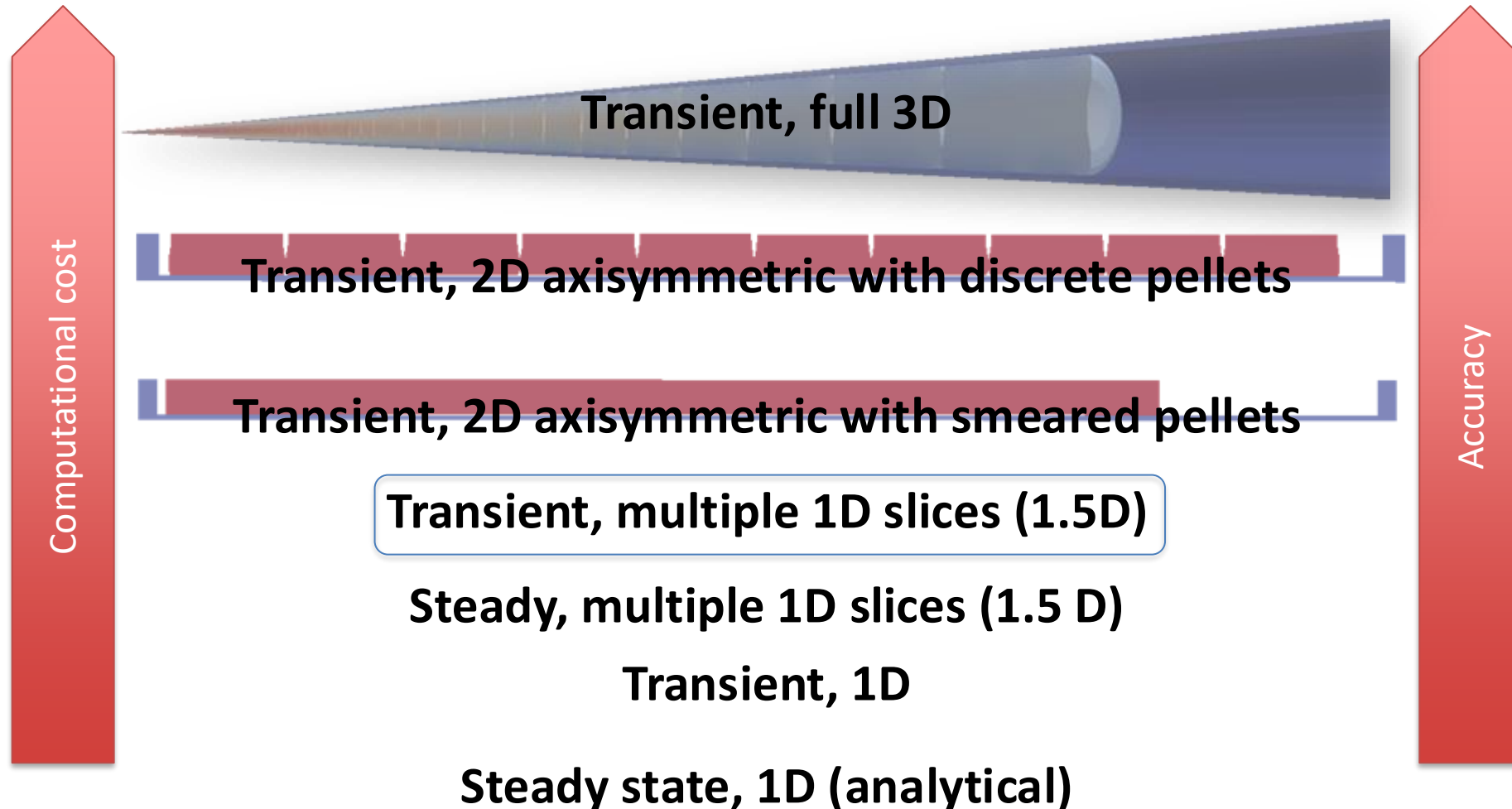


FRAPCON Flow Chart

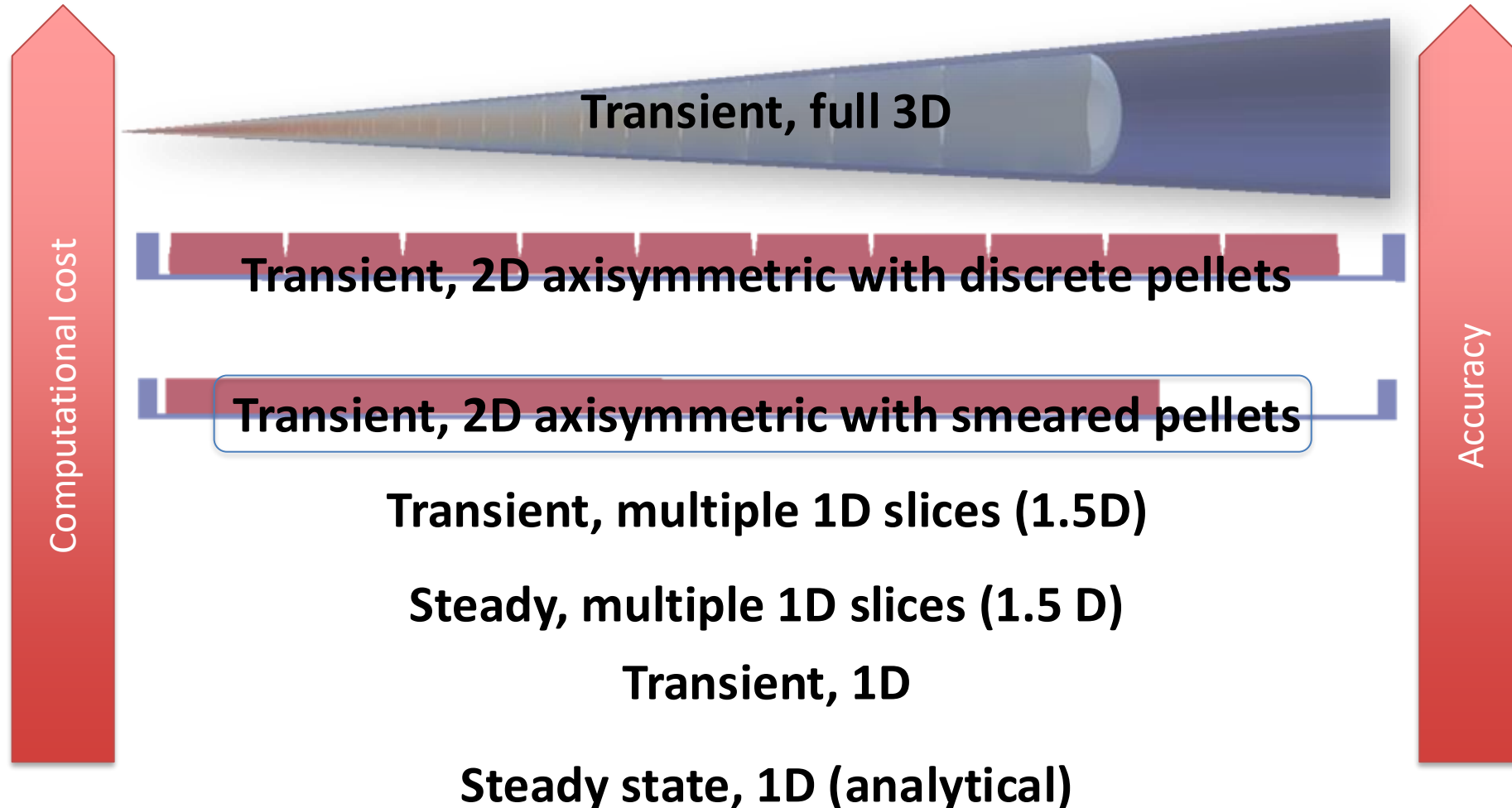
- FRAPCON is the NRC's steady-state fuel performance code
- FRAPCON has the ability to accurately calculate the high-burnup response of light-water reactor fuel rods
- FRAPCON iterates to determine fuel rod temperatures, fuel and cladding deformation
- This converged data is iterated to produce gas release, void volumes and plenum pressure
- Then marches forward in time



FRAPTRAN



FALCON



FALCON

- FALCON is a 2D fuel performance code developed by EPRI
- Development of FALCON started in 1996
- The beta version was released in 2003
- It was developed by ANATECH for EPRI
- FALCON is proprietary, owned by EPRI
- It is no longer under active development in the US

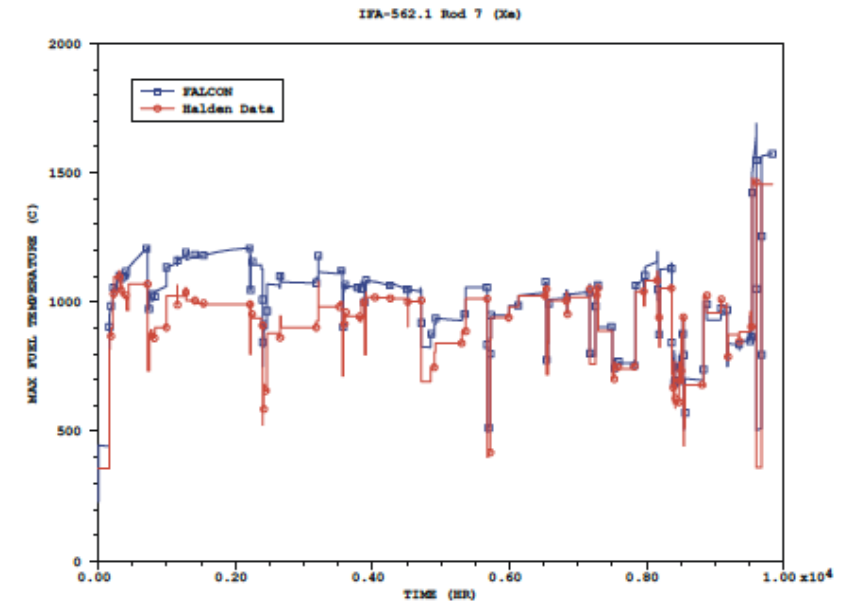


Figure 3-6
Calculated Fuel Temperatures Versus Measured Data for IFA-562.1 Rod 7 (Xe-filled)

FALCON is a 2D transient and steady state code

Fuel

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q$$

Solved with FEM

$$0 = \nabla \cdot \sigma$$

Solved with FEM

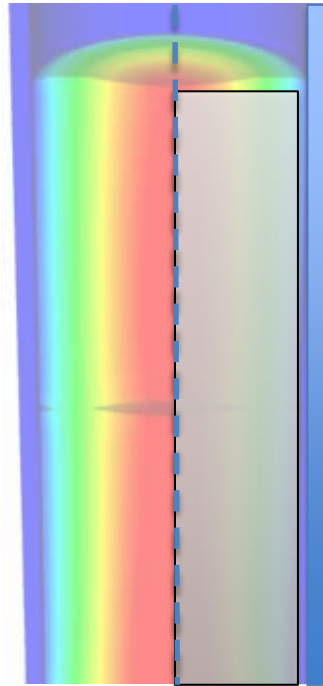
Cladding

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q$$

Solved with FEM

$$0 = \nabla \cdot \sigma$$

Solved with FEM



Gap

- Pressure is calculated using equation of state
- Simplified contact model is used for gap closure
- Gap heat transfer model is used

FALCON can predict the fuel performance in axisymmetric RZ space or in R θ space

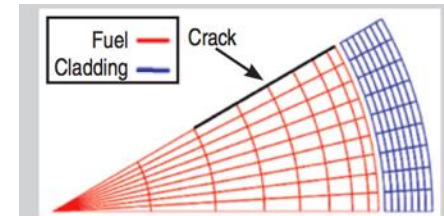
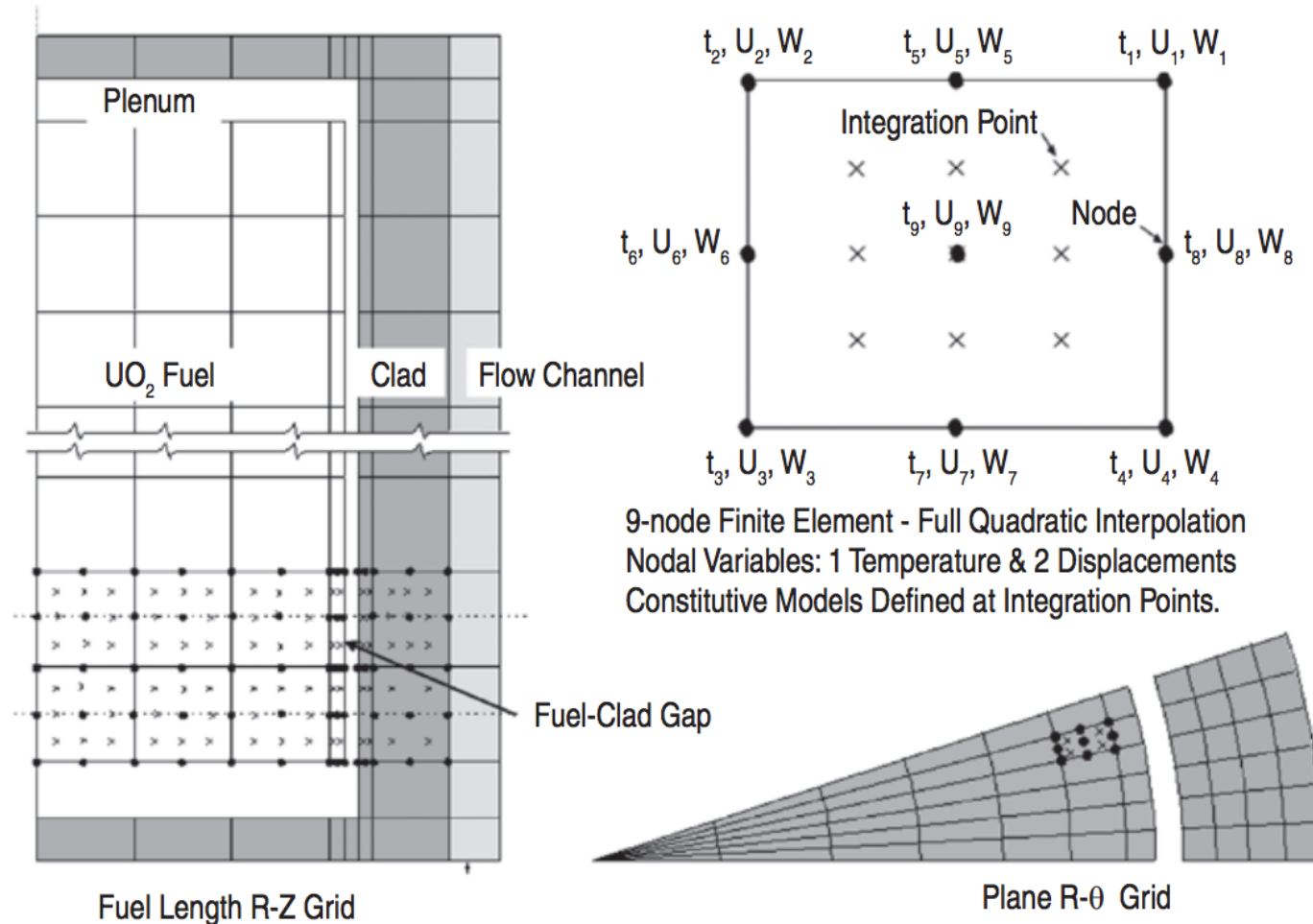


Figure 3. Standard PCI model.

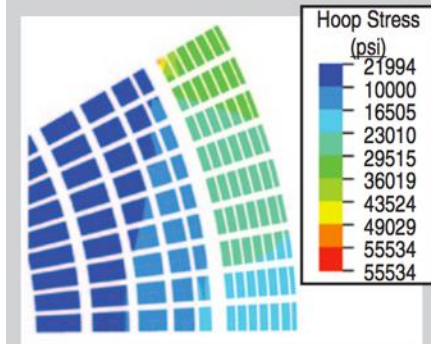


Figure 4. Calculated cladding hoop stress distribution (psi) using the standard PCI model.

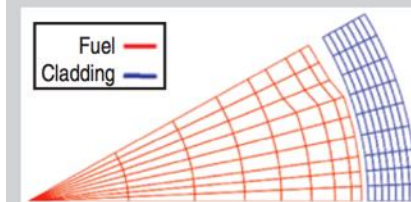
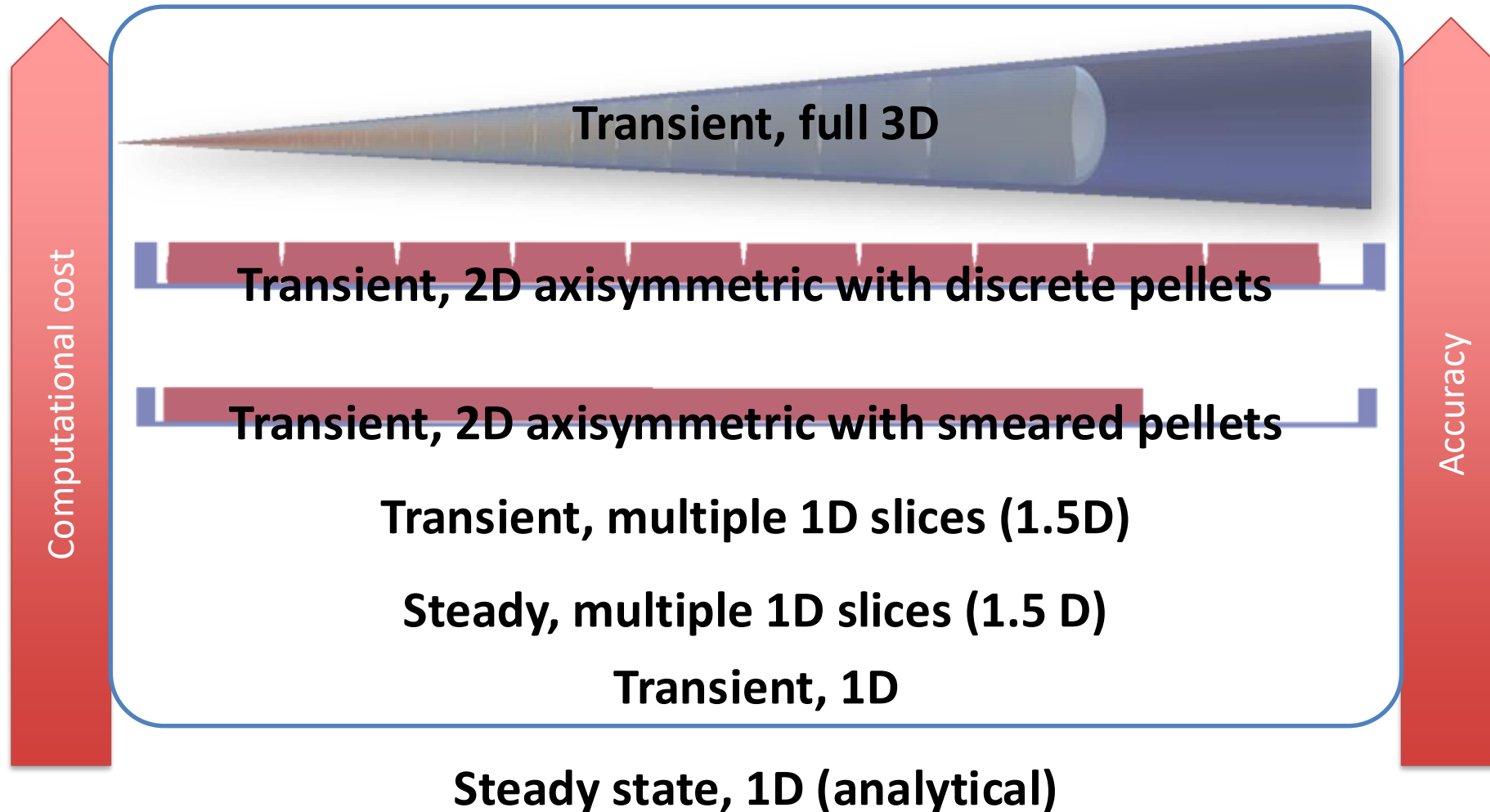


Figure 5. Missing pellet surface (MPS) PCI model.

BISON



BISON

- The next generation fuel performance code under development in the US
- It uses the MOOSE framework
- Development was begun in 2008
- The first paper using BISON was published in 2009 and the paper summarizing its full capabilities was published in 2012
- It was developed at Idaho National Laboratory, with some support by ANATECH
- BISON is available for free, but it is export controlled and requires a license agreement be signed

BISON models the fuel behavior ranging from 1D to full 3D and uses FEM

Fuel

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q$$

Solved with FEM

$$0 = \nabla \cdot \sigma$$

Solved with FEM

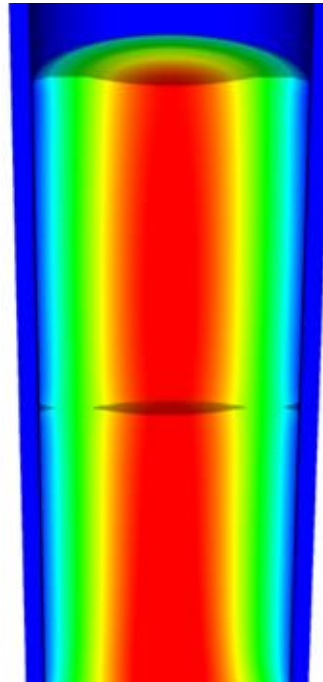
Cladding

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q$$

Solved with FEM

$$0 = \nabla \cdot \sigma$$

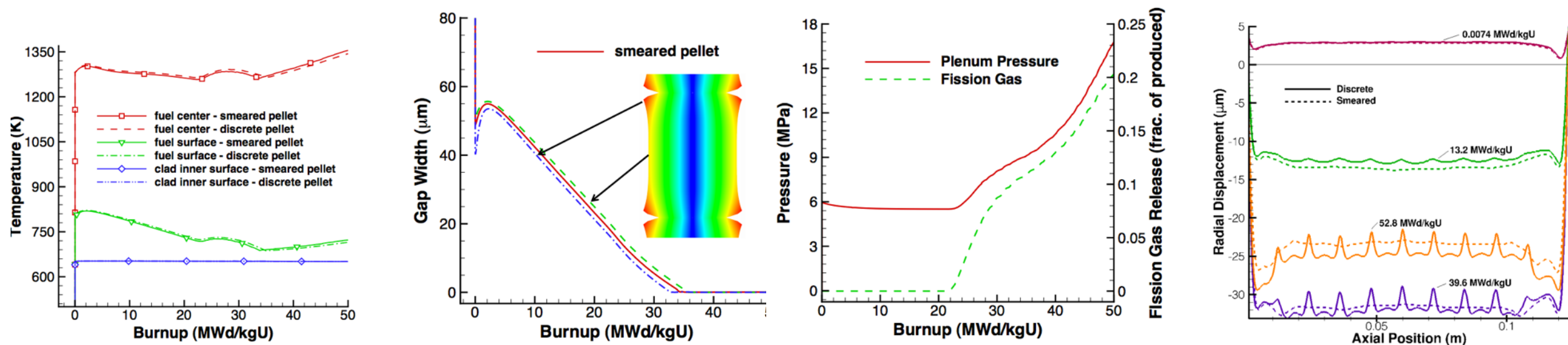
Solved with FEM



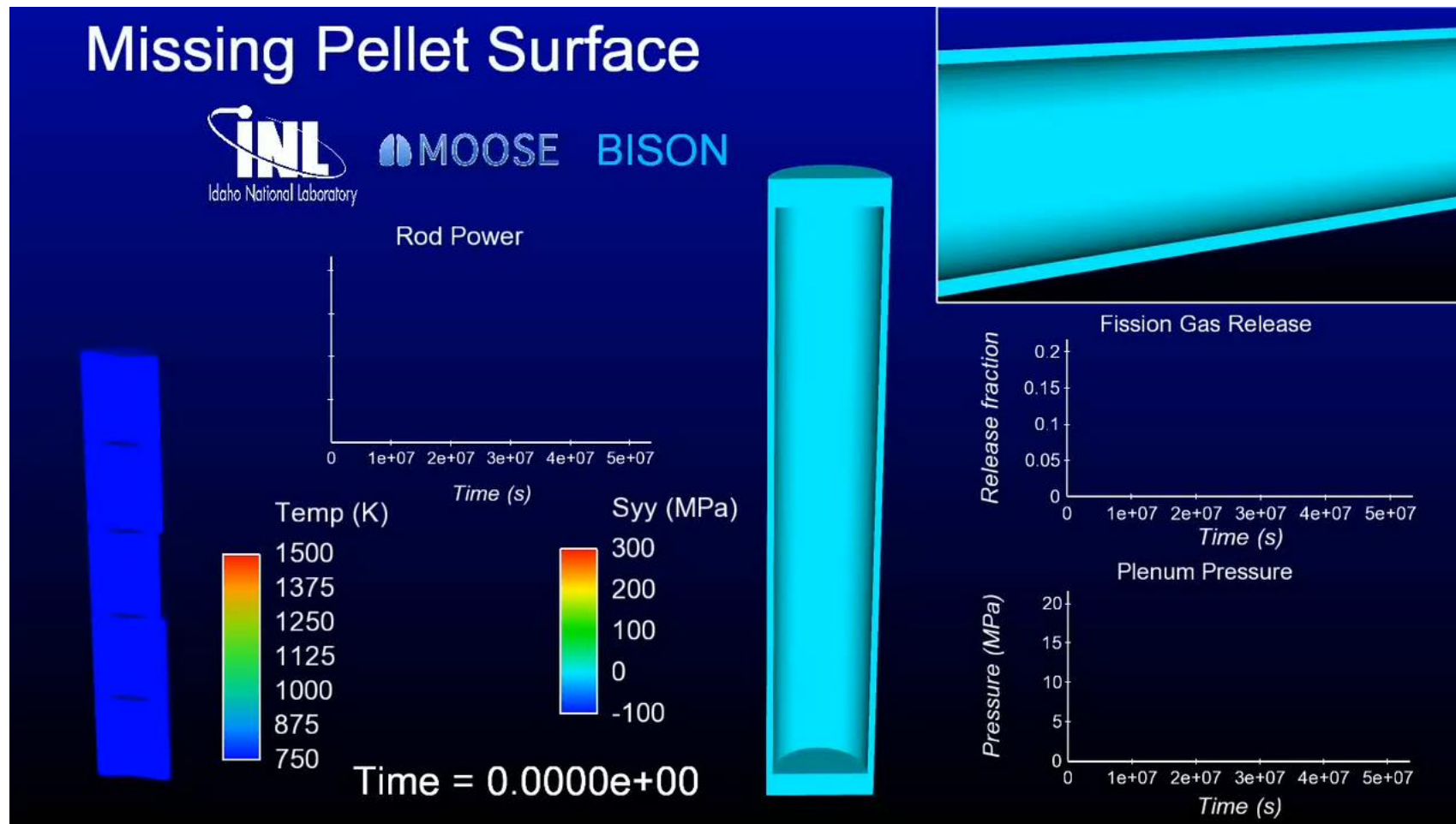
Gap

- Pressure is calculated using equation of state
- Fully implemented implicit contact algorithm
- Gap heat transfer model is used

Can handle smeared or discrete pellets, asymmetric pellet geometry and deformation



Because of its unique 3D capability, BISON can model truly 3D fuel performance problems

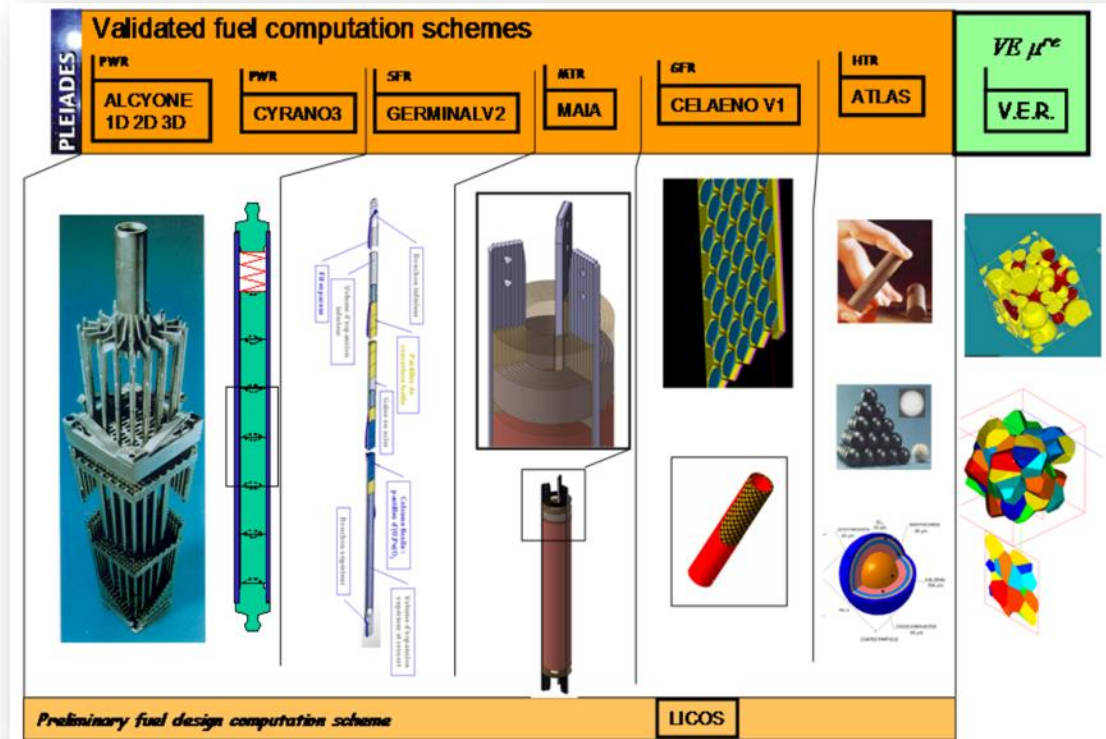


Other countries have other fuel performance codes

Table 1

List of fuel rod performance codes developed in different parts of the world for light water reactor fuel. More information is provided in Appendix: Overview of main fuel rod performance code developments across the globe.

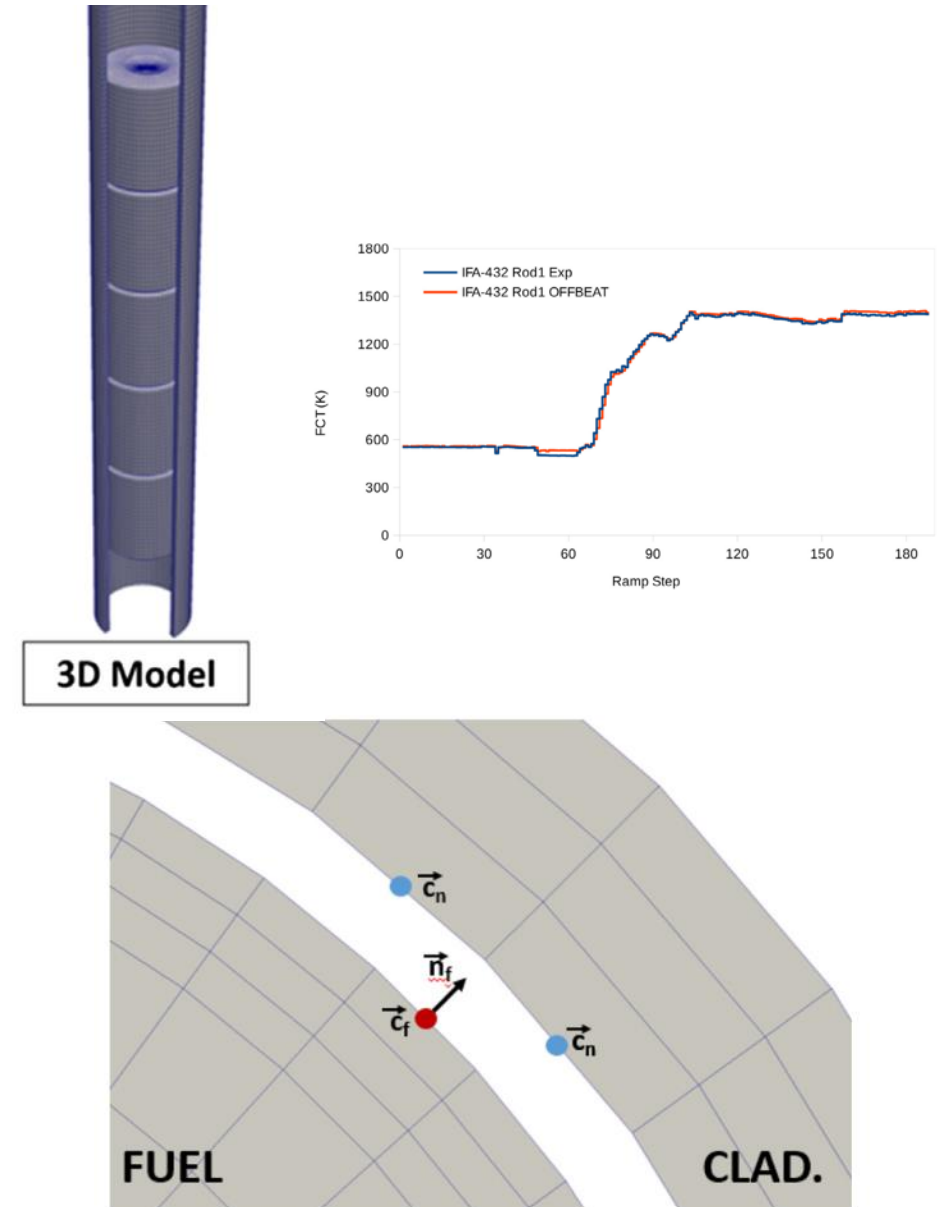
Country	Organization	Code name (precursor codes)
Argentina	CNEA	BACO, DIONISIO
Belgium	Belgonucleaire	COMETHE
	SCK-CEN	MACROS (ASFAD)
China	Xi'an Xiaotong University	FROBA
	CIAE	FTPAC
	NPIC	FUPAC
	CGNPC	JASMINE
Czech Republic	UJV	PIN-MICRO
France	CEA	(GAPCON-THERMAL2)
	Framatome	ALCYONE
		(METEOR-TRANSURANUS)
		COPERNIC (TRANSURANUS),
		GALILEO (COPERNIC/RODEX/CARO)
	EdF	CYRANO
	IRSN	SCANAIR
Germany	Siemens	CARO
	Framatome	GALILEO (COPERNIC/RODEX/CARO)
	GRS	TESPA-ROD (TESPA)
	JRC	TRANSURANUS (URANUS)
Hungary	MTA EK	FUROM (PIN-MICRO)
India	BARC	FAIR, PROFESS
	PNC	FUDA
Japan	CRIEPI	EIMUS (FEMAXI-III)
	JAEA	FEMAXI, RANNS
	SEPC	IRON (FEMAXI-III)
	NFD	TRUST
Korea	KAERI	COSMOS, INFRA
Russian Federation	VNIINM	START, RAPTA
	TRINITI	RTOP
	IBRAE	SFPR (MFPR)
Sweden	Westinghouse	STAV
	Sweden Electric	
United Kingdom	NNL, EDF Energy	ENIGMA (MINIPAT, SLEUTH, HOTROD)
USA	USNRC	FRAPCON, FRAPTRAN (FRAP), FAST
	Siemens	RODEX
	EPRI	FALCON (FREY, ESCORE)
	INL	BISON
	Framatome	GALILEO (COPERNIC/RODEX/CARO)
	Westinghouse	PAD



<http://www.materials.cea.fr/en/PDF/PLEIADES-Platform.pdf>

OFFBEAT – new(ish) tool

- The OpenFOAM Fuel BEhavior Analysis Tool (OFFBEAT) is a multi-dimensional fuel performance code developed in Switzerland
- The code can be used both for studying complex 2D or 3D local effects and for more traditional 1.5D base irradiation analyses
- OFFBEAT is based on the open-source C++ library OpenFOAM, thus the governing equations are discretized with modern finite volume techniques



Summary

- Fuel performance codes are focused on predicting the center temperature of the pellet and the stress in the cladding
- All fuel performance codes
 - Numerically model the temperature in the fuel
 - Numerically model the stress in the cladding
 - Consider gap pressure, closure, and heat transfer
- The primary US codes are
 - FRAPCON – Steady state 1.5D, uses finite difference
 - FRAPTRAN – Transient 1.5D, uses finite difference
 - BISON – Steady or transient, 1D – 3D, uses finite element